Enrico **Magnago**

APPLIED SCIENTIST · PHD · SOFTWARE DEVELOPER

enmag.github.io - EnMag - EnMag - enrico-magnago - EnMag

Inquisitive, self-motivated and perseverant problem solver with outstanding academic results, strong background in formal verification and industrial experience in large-scale distributed systems. Dependable and proficient software developer with experience in design, implementation and evaluation of symbolic model checking algorithms. Love to combine formal verification expertise and coding skills to build scalable verification tools and create reliable software products.

Expertise: Formal Verification, Temporal Logics, SAT, SMT, Timed / Hybrid Systems, Software Testing, Software Development, S3.

Experience

- **SENIOR APPLIED SCIENTIST** — *Berlin, Germany*
  Amazon Web Services — *2024/10/01 - ongoing*
  - Automated API learning, software verification and testing of large-scale distributed systems.
- **APPLIED SCIENTIST II** — *Berlin, Germany*
  Amazon Web Services — *2022/12/01 - 2024/09/30*
  - Automated testing of large-scale distributed systems.
- **RESEARCH AND DEVELOPMENT ENGINEER** — *Dublin, Ireland*
  Synopsys Inc — *2022/05/09 - 2022/10/31*
  - Design and development of tools for **static analysis** of interpreted programming languages.
- **PHD STUDENT / CANDIDATE** — *Trento, Italy*
  University of Trento, Fondazione Bruno Kessler — *2018/11/01 - 2022/11/18*
  - Significantly enhanced the **falsification** capabilities of model checkers in **infinite-state** and **timed systems**.
  - Designed an effective **SMT-based symbolic algorithm** capable of finding counterexamples that other state-of-the-art procedures miss.
  - Developed a novel **compositional** semi-automated approach to falsify temporal properties. item Evaluated the viability of cloud platforms (**AWS**, **Azure**) for performance testing.
  - Published **5 research papers**: 3 in highly rated international conferences and 2 in technical journals.
- **TEACHING ASSISTANT: FORMAL METHODS COURSE, MASTER IN COMPUTER SCIENCE** — *Trento, Italy*
  University of Trento — *2019/03/15 - 2019/12/31*
  - Gave the laboratory lectures, defined the exam exercises and evaluated the students' solutions.
  - Topics: model checking using **Spin**, **NuSMV** and **nuXmv**; verification of timed systems using **timed-nuXmv** and **HyCOMP**.
- **SOFTWARE DEVELOPER** — *Trento, Italy*
  Fondazione Bruno Kessler, Embedded Systems (ES) unit — *2017/03/27 - 2018/10/31*
  - Extended the symbolic model checker nuXmv to support verification of **timed systems**.
  - Designed and implemented techniques to verify expressive specifications: **Metric Temporal Logic**.
  - Defined and evaluated a novel and effective approach to identify infinite **non-lasso counterexamples** for timed systems.
  - Developed a **new compiler module** in nuXmv to meet software re-usability and extensibility requirements.
  - Testing, bug fixing, refactoring and documentation of **large code base**:
    >1M lines of **C**, **>200K** lines of **Python**, **>150K** lines of **C++**, **>50k** lines of **LaTeX**.
- **INTERN** — *Trento, Italy*
  Fondazione Bruno Kessler, Data & Knowledge Management (DKM) unit — *2015/07/20 - 2015/10/20*
  - Implemented a Semantic Web application database interface using **Elasticsearch** Java API.

Education

- **PHD, INFORMATION AND COMMUNICATION TECHNOLOGY** — *Trento, Italy*
  Fondazione Bruno Kessler, University of Trento — *2018/11/01 - 2022/11/18*
  - Cum Laude.
  - Advisor: Alessandro Cimatti, Co-advisor: Alberto Griggio.
- **SUMMER / WINTER SCHOOLS ABOUT FORMAL VERIFICATION** — *Marktoberdorf, Verona, Lisbon*
  Technical University of Munich, University of Verona, University of Lisbon
  - **MOD 2019**: Summer School on Safety and Security of Software Systems: Logics, Proofs, Applications. — *2019/07/31 - 2019/08/09*
  - **CPS 2019**: Summer School on Formal Methods for Cyber-Physical Systems. — *2019/06/03 - 2019/06/07*
  - **VMCAI 2019**: Winter School on Verification, Model Checking, and Abstract Interpretation. — *2019/01/09 - 2019/01/12*
- **MASTER DEGREE IN COMPUTER SCIENCE** — *Trento, Italy*
  University of Trento — *2016/08/08 - 2018/10/10*
  - Final grade: 110L/110, Average grade: 29.24/30.
  - Notable project: distributed consensus protocol in Go.

- **Post-graduate Erasmus student** — *Edinburgh, United Kingdom*
  University of Edinburgh, School of Informatics — *2016/09/05 - 2016/12/24*
  - Notable project: algorithms using Hadoop Map-Reduce.
- **Bachelor Degree in Computer Science** — *Trento, Italy*
  University of Trento — *2013/08/08 - 2016/07/18*
  - Final grade: 110L/110, Average grade: 28.11/30.
  - Notable project: multi-process question-answer game in C.
- **High school diploma** — *Cles, Italy*
  Liceo Bertrand Russell — *2008/09/19 - 2013/07/04*
  - Final grade: 93/100.

Programming Languages

C	●●●●○	NuSMV[1], nuXmv[1].
C++	●●●●○	nuXmv[1], msatic3[1].
Python	●●●●○	F3[2], pySMT[1], General Scripting.
Java	●●●●○	Development of large scale services.
LaTeX	●●●○○	Publications, Tools Documentation, BSc, MSc and PhD Theses.
Lex-Yacc	●●●○○	NuSMV[1], nuXmv[1], Academic Project.
Scala	●●●○○	Professional use.
Bash	●●○○○	General Scripting.
Rust	●○○○○	Self-study.

1: Pre-existent tool or library
2: Standalone software

Skills

Languages:	●●●●●	Italian	●●●●○	English
Hobbies:	●●●○○	Trumpet	●●●○○	Skiing

Publications

INFOCOMP 2022 — **LTL falsification in infinite-state systems**
Alessandro Cimatti, Alberto Griggio, and Enrico Magnago.
Information and Computation Journal, Volume 289.

ATVA 2021 — **Automatic Discovery of Fair Paths in Infinite-State Transition Systems**
Alessandro Cimatti, Alberto Griggio, and Enrico Magnago.
19^{th} *International Symposium on Automated Technology for Verification and Analysis.*

VMCAI 2021 — **Proving the Existence of Fair Paths in Infinite-State Systems**
Alessandro Cimatti, Alberto Griggio, and Enrico Magnago.
22^{th} *International Conference on Verification, Model Checking, and Abstract Interpretation.*

INFOCOMP 2020 — **SMT-based satisfiability of first-order LTL with event freezing functions and metric operators**
Alessandro Cimatti, Alberto Griggio, Enrico Magnago, Marco Roveri, and Stefano Tonetta.
Information and Computation Journal, Volume 272.

CAV 2019 — **Extending nuXmv with Timed Transition Systems and Timed Temporal Properties**
Alessandro Cimatti, Alberto Griggio, Enrico Magnago, Marco Roveri, and Stefano Tonetta.
31^{st} *International Conference on Computer Aided Verification.*

Authors are in alphabetical order.

Professional Activities

2025 — CAV Program Committee.
International Conference on Computer Aided Verification.

2025 — FMCAD Program Committee.
International Conference on Formal Methods in Computer-Aided Design.

2025 — OOPSLA Artifact Evaluation Committee.
Conference on Object-Oriented Programming Systems, Languages, and Applications.

2020, 2022 - 2025 — CAV Artifact Evaluation Committee.
International Conference on Computer Aided Verification.

2024 — PLDI Artifact Evaluation Committee.
Symposium on Programming Language Design and Implementation.

2024 — ATVA Artifact Evaluation Committee.
International Symposium on Automated Technology for Verification and Analysis.

2022 — FORMATS Artifact Evaluation Committee.
International Conference on Formal Modeling and Analysis of Timed Systems.